PREDICTING CONTINUOUS TARGET VARIABLES WITH REGRESSION ANALYSIS - WEEK 10

- REGRESSION MODELS - PREDICT TARGET VARIABLES ON A CONTINUOUS SCALE → ATTRACTIVE FOR SCIENCE QUESTIONS.
  ↳ OTHER EXAMPLES = UNDERSTANDING RELATIONSHIP WITH VARIABLES, EVALUATING TRENDS, MAKING FORECASTS.

LINEAR REGRESSION

- LINEAR REGRESSION - MODEL RELATIONSHIP BETWEEN ONE OR MULTIPLE FEATURES + CONTINUOUS VARIABLES
  ↳ PREDICT OUTPUTS BASED ON CONTINUOUS SCALE.

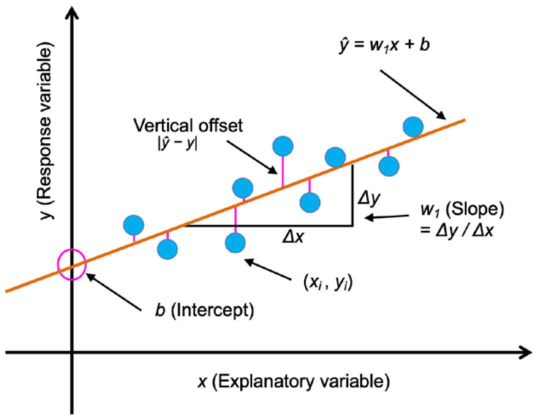
SIMPLE LINEAR REGRESSION

- SIMPLE (UNIVARIATE) LINEAR REGRESSION - MODEL RELATIONSHIP BETWEEN (EXPLANATORY VARIABLE, x) + CONTINUOUS TARGET (RESPONSE VARIABLE, y)

$y = mx + b$ → HE USES - $y = w_1 x + b$

b = BIAS UNIT - REPRESENTS y AXIS INTERCEPT
$m(w_1)$ = WEIGHT COEFFICIENT

- GOAL → LEARN WEIGHT OF LINEAR EQUATION TO DESCRIBE RELATION BETWEEN x, y → TO PREDICT NEW EXPLANATORY VARIABLE NOT PART OF TRAINING DATASET.

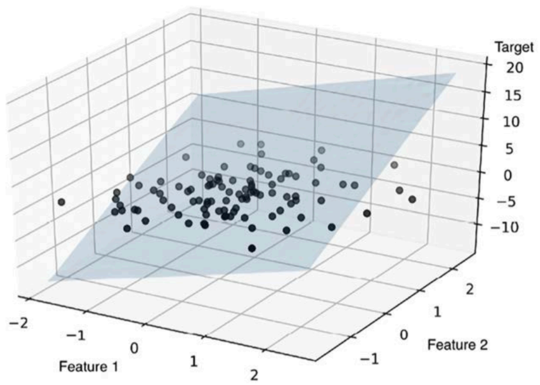


- BEST FITTING LINE → REGRESSION LINE
- VERTICAL LINE OFF REGRESSION LINE → OFFSET/RESIDUAL - ERRORS OF PREDICTIONS.

MULTIPLE LINEAR REGRESSION

- MULTIPLE LINEAR REGRESSION - GENERALISED LINEAR REGRESSION TO MULTIPLE EXPLANATORY VARIABLES

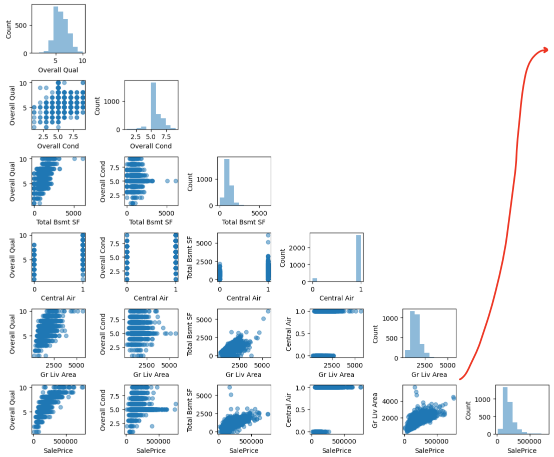
$$y = w_1 x_1 + \dots + w_m x_m + b = \sum_{i=1}^{m} w_i x_i + b = w^T x + b$$



- MULTIPLE LINEAR REGRESSION → FITS TO DATASET WITH THREE OR MORE FEATURES
- SIMPLE + MULTIPLE LINEAR REGRESSION → BASED ON THE SAME CONCEPTS + SAME EVALUATION TECHNIQUE

VISUALISING IMPORTANT CHARACTERISTIC OF A DATASET

- EXPLANATORY DATA ANALYSIS (EDA) - IMPORTANT + RECOMMENDED FIRST STEP BEFORE TRAINING ML MODEL.
- SCATTERPLOT MATRIX → VISUALISE PAIR-WISE CORRELATION BETWEEN DIFFERENT FEATURES IN DATASET IN ONE PLACE.



- SCATTERPLOT MATRIX - QUICKLY SEE HOW DATA IS DISTRIBUTED + WHETHER IT CONTAINS OUTLIERS.
- 'Gr Liv Area' AND 'SalePrice' → LINEAR RELATIONSHIP
- TRAINING A LINEAR REGRESSION MODEL → DOESN'T REQUIRE NORMALLY DISTRIBUTED VARIABLES (CAUGHT)

RELATIONSHIPS USING CORRELATION MATRIX

- CORRELATION MATRIX - QUANTIFY + SUMMARISE LINEAR RELATIONSHIPS BETWEEN VARIABLES
  - ↳ CLOSELY RELATED TO COVARIANCE MATRIX
- CORRELATION MATRIX → RESCALED VERSION OF COVARIANCE MATRIX
- CORRELATION MATRIX = COVARIANCE MATRIX FROM STANDARDISED FEATURES.
- CONTAINS PEARSONS PRODUCT - MOMENT CORRELATION COEFFICIENT (PEARSON'S R) → MEASURES LINEAR DEPENDANCE BETWEEN PAIRS OF FEATURES.
  - ↳ RANGE -1 TO 1.
  - ↳ r = 1 , r = 0 = PERFECT POSITIVE CORRELATION , r = -1 PERFECT NEGATIVE CORRELATION

$$r = \frac{\sum_{i=1}^{n}\left[(x^{(i)}-\mu_x)(y^{(i)}-\mu_y)\right]}{\sqrt{\sum_{i=1}^{n}(x^{(i)}-\mu_x)^2}\sqrt{\sum_{i=1}^{n}(y^{(i)}-\mu_y)^2}} = \frac{\sigma_{xy}}{\sigma_x\sigma_y}$$

μ = CORRESPONDING FEATURE
σ_{xy} = COVARIANCE BETWEEN x AND y
$\sigma_x \sigma_y$ = FEATURES' STANDARD DEVIATION

COVARIANCE VS. CORRELATION FOR STANDARDISED FEATURES

- STANDARDISE FEATURE x AND y.

$$x' = \frac{x-\mu_x}{\sigma_x}, \quad y' = \frac{y-\mu_y}{\sigma_y}$$

- COMPUTE (POPULATION) COVARIANCE BETWEEN FEATURES:

$$\sigma_{xy} = \frac{1}{n}\sum_{i}^{n}(x^{(i)}-\mu_x)(y^{(i)}-\mu_y)$$

- STANDARDISED FEATURES = MEAN ZERO → CALCULATE COVARIANCE:

$$\sigma'_{xy} = \frac{1}{n}\sum_{i}^{n}(x'^{(i)}-0)(y'^{(i)}-0)$$

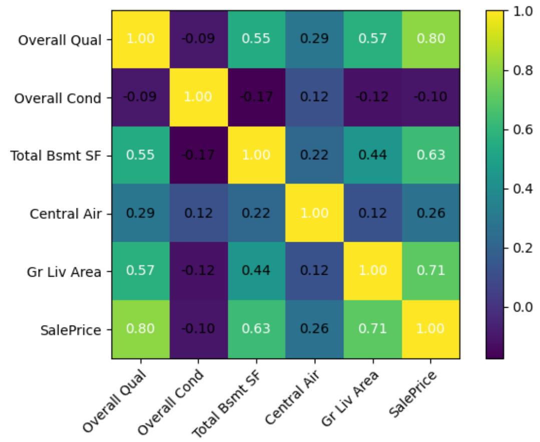
- RESUBSTITUTION:

$$\sigma'_{xy} = \frac{1}{n}\sum_{i}^{n}\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)$$

$$\sigma'_{xy} = \frac{1}{n\cdot\sigma_x\sigma_y}\sum_{i}^{n}(x^{(i)}-\mu_x)(y^{(i)}-\mu_y)$$

- SIMPLIFY:

$$\sigma'_{xy} = \frac{\sigma_{xy}}{\sigma_x\sigma_y}$$



- GIVES US INFORMATION ABOUT LINEAR CORRELATION

ORDINARY LEAST SQUARES LINEAR REGRESSION MODEL

- ORDINARY LEAST SQUARES (OLS) - SOMETIMES CALLED LINEAR LEAST SQUARES - ESTIMATE PARAMETERS OF LINEAR REGRESSION LINE TO MINIMISE ERRORS

REGRESSION PARAMETERS WITH SGD

- LOSS FUNCTION OF ADALINE = MEAN SQUARE ERROR (MSE) → SAME FOR OLS.

$$L(w,b) = \frac{1}{2n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2$$

$\hat{y}$ = PREDICTED VALUE $\hat{y} = w^T x + b$

- OLS → CAN BE UNDERSTOOD AS ADALINE WITHOUT THRESHOLD FUNCTION → SO WE CAN GET CONTINUOUS VALUE AND NOT CLASS LABEL (0,1)
- CLOSED - FORM SOLUTION FOR SOLVING OLS:

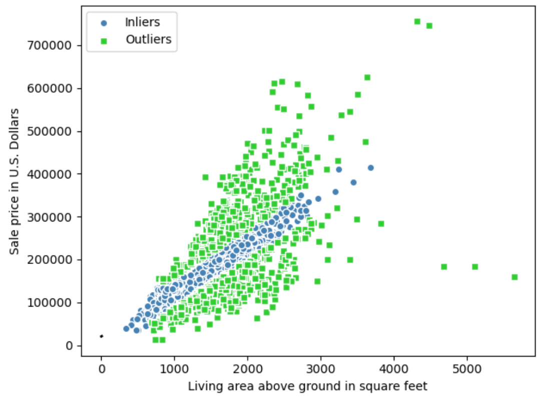
$$w = (X^T X)^{-1} X^T y$$

ROBUST REGRESSION MODEL USING RANSAC

- LINEAR REGRESSION MODELS → CAN BE HEAVILY IMPACTED BY OUTLIERS.
  - ↳ SMALL SUBSET CAN HAVE A BIG EFFECT ON COEFFICIENT.
- RANDOM SAMPLE CONSENSUS (RANSAC) - FITS A REGRESSION MODEL TO SUBSET, CALLED INLIERS
- SUMMARISED ALGO:

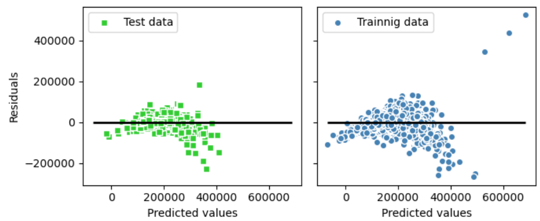
1. SELECT RANDOM NUMBER OF EXAMPLES TO BE INLIERS + FIT THE MODEL.
2. TEST DATA POINTS AGAINST FITTED MODEL + ADD POINTS THAT FALL WITHIN USER-GIVEN TOLERANCE TO INLIERS.
3. REFIT MODEL WITH INLIERS
4. ESTIMATE ERROR OF FITTED MODEL VS. INLIERS
5. TERMINATE ALGO IF PERFORMANCE MEETS CERTAIN USER-DEFINED THRESHOLD, OR FIXED ITERATION NUMBER REACHED, OTHERWISE BACK TO STEP 1.

- USING RANSAC WE REDUCE POTENTIAL OUTLIERS. ONLY ISSUE IS THAT WE DON'T KNOW WHETHER IS POSITIVE OR NEGATIVE EFFECT.



PERFORMANCE EVALUATION OF LINEAR REGRESSION MODELS

- FIRST WE WANT TO SPLIT DATASET TO TRAIN/TEST SPLIT
- RESIDUAL PLOTS - COMMONLY USED GRAPHICAL TOOL FOR REGRESSION MODELS.
  - ↳ THEY CAN DETECT NONLINEARITY AND OUTLIERS + CHECK WHETHER ERRORS ARE RANDOMLY DISTRIBUTED



→ PERFECT PREDICTION → RESIDUAL = 0 (WOULD NEVER HAPPEN)
- GOOD REGRESSION MODEL - ERRORS RANDOMLY DISTRIBUTED + RESIDUALS TO SCATTERED AROUND CENTRE LINE
- PATTERNS IN RESIDUAL MODEL - NOT GOOD - MODEL UNABLE TO CAPTURE SOME EXPLANATORY INFORMATION
- RESIDUAL PLOTS ALSO GOOD FOR DETECTING OUTLIERS

- MSE (MEAN SQUARED ERROR) ALSO GOOD FOR MEASURING MODEL PERFORMANCE. → LOSS FUNCTION TO FIT LINEAR REGRESSION MODEL.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2$$

WITHOUT $\frac{1}{2}$ SCALING FACTOR → USED FOR SIMPLIFYING DERIVATIVE IN GRADIENT DESCENT.

- MSE → NORMALISES ACCORDING TO SAMPLE SIZE, n → MAKES IT POSSIBLE TO COMPARE ACROSS DIFFERENT SAMPLE SIZES.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y^{(i)} - \hat{y}^{(i)}\right|$$

- MAE USED BECAUSE MORE INTUITIVE TO SHOW ERROR ON ORIGINAL UNIT SCALE → EMPHASISES INCORRECT PREDICTIONS SLIGHTLY LESS.
- MSE / MAE FOR COMPARING MODELS → UNBOUND IN CONTRAST TO CLASSIFICATION ACCURACY. → MSE/MAE DEPEND ON DATASET + FEATURE SCALING
- SOMETIMES MORE USEFUL TO REPORT COEFFICIENT OF DETERMINATION (R^2) → STANDARDISED VERSION OF MSE (BETTER INTERPRETABILITY OF MODEL'S PERFORMANCE)
  ↳ R^2 = FRACTION OF RESPONSE VARIANCE

$$R^2 = 1 - \frac{SSE}{SST}$$

SSE = SUM OF SQUARED ERRORS → SIMILAR TO MSE, WITHOUT NORMALISATION BY n

$$SSE = \sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2$$

SST = TOTAL SUM OF SQUARES → VARIANCE OF RESPONSE

$$SST = \sum_{i=1}^{n}\left(y^{(i)} - \mu_y\right)^2$$

R^2 → IS A RESCALED VERSION OF MSE:

$$R^2 = 1 - \frac{SSE}{SST}$$

$$= \frac{\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2}{\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \mu_y\right)^2}$$

$$= 1 - \frac{MSE}{Var(y)}$$

- R^2, BOUNDED BETWEEN 0, 1 → CAN BECOME NEGATIVE FOR TEST DATASET.
- NEGATIVE R^2 = REGRESSION MODEL FITS DATA WORSE THAN HORIZONTAL LINE (SAMPLE MEAN)
- $R^2 = 1$, $MSE = 0$ ⇒ PERFECT FIT

REGULARISED METHODS FOR REGRESSION

- REGULARISATION → TACKLING OVERFITTING, BY ADDING ADDITIONAL INFORMATION → SHRINKS PARAMETER VALUES OF MODEL
- MOST POPULAR REGULARISED LINEAR REGRESSION →
  - RIDGE REGRESSION
  - LEAST ABSOLUTE SHRINKAGE AND SELECTION OPERATOR (LASSO)
  - ELASTIC NET
- RIDGE REGRESSION → L2 PENALISED MODEL → ADD SQUARED SUM OF WEIGHTS TO MSE LOSS FUNCTION:

$$L(w)_{Ridge} = \sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2 + \lambda\|w\|_2^2$$

→ L2 TERM: $\lambda\|w\|_2^2 = \lambda\sum_{j=1}^{m} w_j^2$

- λ - HYPERPARAMETER VALUE → INCREASES REGULARISATION STRENGTH, SHRINKS WEIGHT

- LASSO → CAN LEAD TO SPARSE MODELS → DEPENDING ON REGULARISATION STRENGTH, WEIGHT CAN BECOME ZERO → USEFUL FOR SUPERVISED FEATURE SELECTION:

$$L(w)_{Lasso} = \sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2 + \lambda\|w\|_1$$

→ L1 TERM: $\lambda\|w\|_1 = \lambda\sum_{j=1}^{m}|w_j|$

↳ SUM OF ABSOLUTE MAGNITUDE OF MODEL WEIGHT.

- LIMITATION → LASSO SELECTS AT MOST n FEATURES IF $m > n$, n = NUMBER OF TRAINING EXAMPLES
- PROPERTY OF LASSO → ADVANTAGE - AVOIDS SATURATED MODELS.
- SATURATED MODEL → # OF TRAINING EXAMPLES = # OF FEATURE → FORM OF OVERPARAMETERISATION → TRAINING DATA ALWAYS FIT PERFECTLY.

- ELASTIC NET - L1 PENALTY TO GENERATE SPARSITY + L2 PENALTY (SELECT MORE THAN n FEATURES IF m>n)

$$L(\omega)_{\text{Elastic Net}} = \sum_{i=1}^{n} \left(y^{(i)} - \hat{y}^{(i)}\right)^2 + \lambda_2 \|w\|_2^2 + \lambda_1 \|w\|_1$$

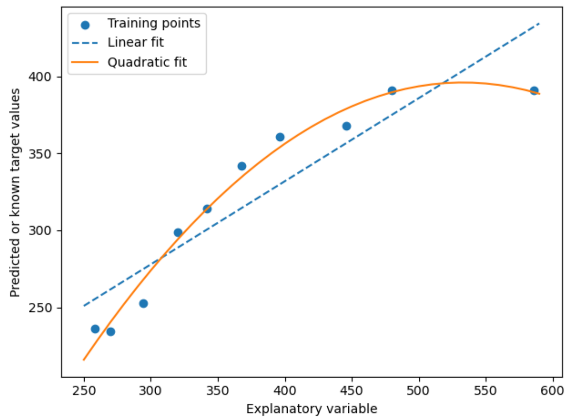
LINEAR REGRESSION MODEL TO CURVE-POLYNOMIAL REGRESSION

- LINEAR + POLYNOMIAL TERMS:

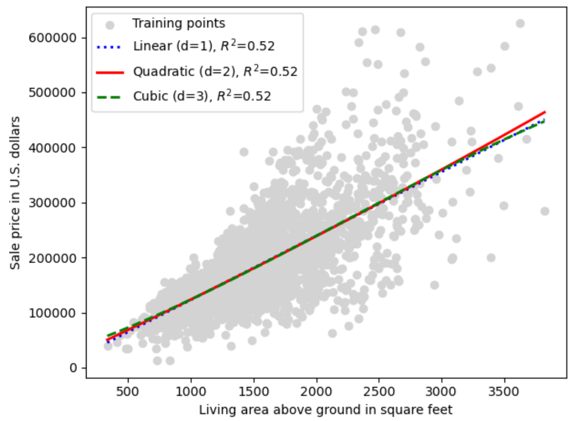
$$y = w_1 x + w_2 x^2 + \dots + w_d x^d + b$$

d = DEGREE OF POLYNOMIAL

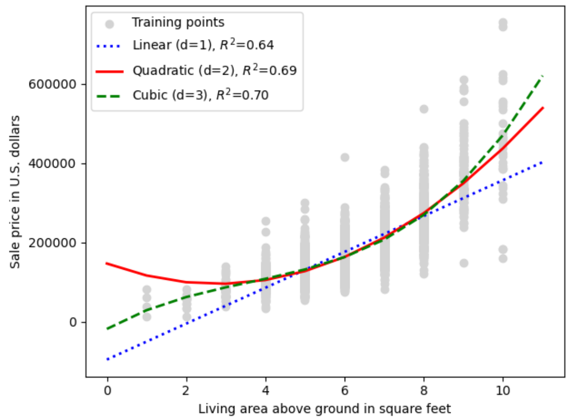
- POLYNOMIAL REGRESSION IS MULTIPLE LINEAR REGRESSION, BECAUSE OF ω



- COMPARISON OF LINEAR VS. QUADRATIC MODEL



- COMPARISON OF DIFFERENT CURVES FITTED TO SALE PRICE/LIVING AREA DATA
- QUADRATIC + CUBIC DON'T HAVE EFFECT → RELATIONSHIP IS LINEAR



- LINEAR, QUADRATIC + CUBIC FIT
- QUADRATIC + CUBIC BETTER CAPTURE RELATIONSHIP
- ADDING MORE POLYNOMIAL FEATURES → INCREASES COMPLEXITY → INCREASES OVERFITTING CHANCES
- IN PRACTICE → TEST ON SEPARATE TEST DATASET.

NONLINEAR RELATIONSHIPS USING RANDOM FORESTS

DECISION TREE REGRESSION

- ADVANTAGE → WORKS WITH ARBITRARY FEATURES + DOESN'T REQUIRE ANY TRANSFORMATION OF FEATURES, IF DEALING WITH NONLINEAR DATA.
- DEFINED ENTROPY AS MEASURE OF IMPURITY TO DETERMINE WHICH FEATURE SPLIT MAXIMISES INFORMATION GAIN (IG).

$$IG(D_p, x_i) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

x_i = FEATURE THAT PERFORMS SPLIT
D_p = SUBSET OF TRAINING EXAMPLES AT PARENT NODE
N_p = # OF TRAINING EXAMPLES IN PARENT NODE
D_{left}, D_{right} = SUBSETS OF TRAINING EXAMPLES AFTER SPLIT
I = IMPURITY FUNCTION

- GOAL: FIND FEATURE SPLIT THAT MAXIMISES INFORMATION GAIN
- IMPURITY METRIC FOR CONTINUOUS VARIABLES → IMPURITY MEASURE OF A NODE, t, AS MSE:

$$I(t) = MSE(t) = \frac{1}{N_t} \sum_{i \in D_t} \left(y^{(i)} - \hat{y}_t\right)^2$$

N_t = # OF TRAINING EXAMPLES AT NODE, t
D_t = TRAINING SUBSET AT NODE, t
$y^{(i)}$ = TRUE TARGET VALUE
$\hat{y}_t$ = PREDICTED VALUE

$$\hat{y}_t = \frac{1}{N_t} \sum_{i \in D_t} y^{(i)}$$

- CONTEXT OF DECISION TREES → MSE REFERRED AS WITHIN-NODE VARIANCE → SPLITTING CRITERION BETTER KNOWN AS VARIANCE REDUCTION.

RANDOM FOREST REGRESSION

- BETTER GENERALISATION PERFORMANCE THAN INDIVIDUAL DECISION TREE BECAUSE OF RANDOMNESS → HELPS DECREASE MODEL'S VARIANCE.
- LESS SENSITIVE TO OUTLIERS + DOESN'T REQUIRE HYPERPARAMETER TUNING → ONLY TUNING REQUIRED = NUMBER OF TREES IN ENSEMBLE
- RANDOM FOREST ALGORITHM - BASICALLY THE SAME AS BASIC RANDOM FOREST ALGORITHM
  ↳ DIFFERENCE = MSE USED TO GROW INDIVIDUAL DECISION TREES + PREDICT TARGET VARIABLES CALCULATED AS AVERAGE PREDICTION ACROSS ALL DECISION TREES.